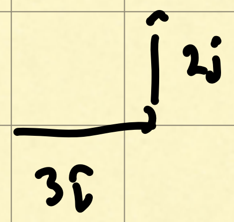
#13: Change of Basis

$\begin{bmatrix}3\\2\end{bmatrix} = \begin{matrix}3\cdot i\\2\cdot j\end{matrix}$



choice of i & j

affects what scalar coords scale

- coord sys = how to translate b/w vectors & coords
- what if diff set of basis vectors?
- how to translate b/w coordinate systems?

new basis vectors $b_1\begin{bmatrix}2\\1\end{bmatrix}$ $b_2\begin{bmatrix}-1\\1\end{bmatrix}$ for vector $\begin{bmatrix}-1\\2\end{bmatrix}$: $-1\begin{bmatrix}2\\1\end{bmatrix}+2\begin{bmatrix}-1\\1\end{bmatrix}$

$= \begin{bmatrix}-4\\1\end{bmatrix}$ in standard bases

- matrix-vector multiplication w/ col

= Jenn's new basis vectors.

$$\begin{array}{cc} b_1 & b_2 \end{array}$$

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix} \text{ in our land}$$

new basis vectors

her vector

↳ transform basis

transforms

our grid → Jenn's grid

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$$

ours ← Jenn's lang

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

change of basis

inverse change of basis matrix

$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \longrightarrow$ in her lang? : $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

our lang

$= \begin{bmatrix} 5/3 \\ 1/3 \end{bmatrix}$

same vector in her language

$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$

Jenn's basis vectors written in our coords

$A \begin{bmatrix} x_j \\ y_j \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$

v in her coords

v in our coords

$\begin{bmatrix} x_j \\ y_j \end{bmatrix} = A^{-1} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$

- But we also represent transformations w/ coordinates.

- transformation matrices?

- [change of basis matrix] [vector in Jen's lang] = vector in our language

  ... = trans. matrix in Jen's lang

[transformation in our lang] · [ ] [ ] = transformed vector in our lang

[inverse change of basis matrix]$^{-1}$ · = transformed vector in her language

$A^{-1} M A$ = math empathy

M = transformation as you see it

outer two = empathy, shift in perspective

full product = same transformation but as someone else sees it

we care ab alternate coordinate systems bc ....